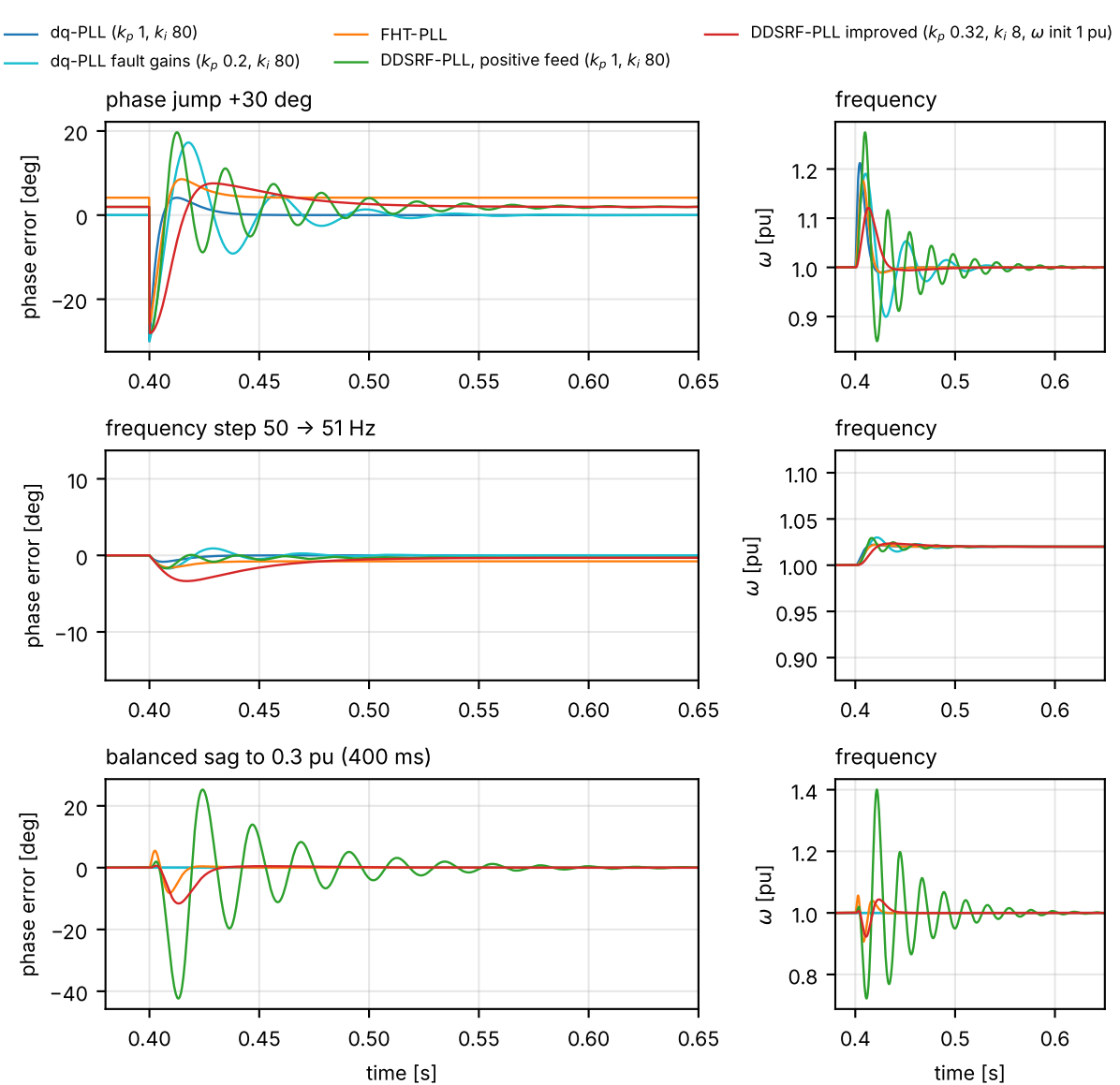
dq-PLL (k_p 1, k_i 80)
dq-PLL fault gains (k_p 0.2, k_i 80)
FHT-PLL
DDSRF-PLL, positive feed (k_p 1, k_i 80)
DDSRF-PLL improved (k_p 0.32, k_i 8, ω init 1 pu)
phase jump +30 deg
frequency
frequency step 50 → 51 Hz
frequency
balanced sag to 0.3 pu (400 ms)
frequency
phase error [deg]
ω [pu]
time [s]